



**Master of Computer Application – Game Development
Semester – I**

Sub: Front-End Frameworks and Technologies

Topic: JavaScript Grade Calculator

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October 2026

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INTRODUCTION

The JavaScript Grade Calculator is an interactive, client-side web application developed using semantic HTML5 and vanilla JavaScript (ES5/ES6). Built to streamline academic evaluation, the tool automates the computation of aggregate marks, percentage ratios, and standardized letter grades from subject-wise numerical inputs.

By leveraging HTML5 form constraints (`type="number"`, `min`, `max`, and `required`), the application establishes baseline client-side data validation directly at the input level. JavaScript handles DOM event management via the `onsubmit` handler, executing arithmetic evaluations, conditional grading ladders (if-else branching), and per-subject threshold checks to ascertain pass/fail status. Dynamic UI manipulation using DOM property binding (`innerHTML`) and visibility toggling (`hidden`) ensures that calculated metrics are instantly presented to the user without triggering page reloads. This project showcases the foundational mechanics of JavaScript-driven DOM manipulation, dynamic data rendering, and event-driven front-end programming.

INPUT CODE

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <title>JavaScript Grade Calculator</title>
  </head>
  <body>
    <h1>Grade Calculator</h1>

    <form id="gradeForm">
      <label>Student Name: <input type="text" id="studentName"
required></label>
      <br><br>
      <label>English: <input type="number" id="english" min="0" max="100"
required></label>
      <br><br>
      <label>Mathematics: <input type="number" id="maths" min="0" max="100"
required></label>
      <br><br>
      <label>Science: <input type="number" id="science" min="0" max="100"
required></label>
      <br><br>
      <label>Computer: <input type="number" id="computer" min="0"
max="100" required></label>
      <br><br>
      <label>Social Science: <input type="number" id="socialScience" min="0"
max="100" required></label>
      <br><br>
      <button type="submit">Calculate Grade</button>
    </form>
```

```

<section id="result" hidden>
  <h2>Result</h2>
  <p>Student: <span id="resultName"></span></p>
  <p>Total Marks: <span id="total"></span> / 500</p>
  <p>Average: <span id="average"></span></p>
  <p>Percentage: <span id="percentage"></span>%</p>
  <p>Grade: <span id="grade"></span></p>
  <p>Result: <span id="status"></span></p>
</section>

```

```

<script>
document.getElementById("gradeForm").onsubmit = function () {
  var name = document.getElementById("studentName").value;
  var english = Number(document.getElementById("english").value);
  var maths = Number(document.getElementById("maths").value);
  var science = Number(document.getElementById("science").value);
  var computer = Number(document.getElementById("computer").value);
                                var          socialScience          =
Number(document.getElementById("socialScience").value);
  var total = english + maths + science + computer + socialScience;
  var average = total / 5;
  var grade, status;

  if (average >= 90) grade = "A+";
  else if (average >= 80) grade = "A";
  else if (average >= 70) grade = "B";
  else if (average >= 60) grade = "C";
  else if (average >= 50) grade = "D";
  else grade = "F";

  if (english >= 35 && maths >= 35 && science >= 35 &&
    computer >= 35 && socialScience >= 35) {
    status = "Pass";
  } else {
    status = "Fail";
  }
}

```

```
document.getElementById("resultName").innerHTML = name;
document.getElementById("total").innerHTML = total;
document.getElementById("average").innerHTML = average.toFixed(2);
    document.getElementById("percentage").innerHTML    =
average.toFixed(2);
    document.getElementById("grade").innerHTML = grade;
    document.getElementById("status").innerHTML = status;
    document.getElementById("result").hidden = false;
    return false;
};
</script>
</body>
</html>
```

OUTPUT

Grade Calculator

Student Name:

English:

Mathematics:

Science:

Computer:

Social Science:

Result

Student: Shaurya

Total Marks: 400 / 500

Average: 80.00

Percentage: 80.00%

Grade: A

Result: Pass

CONCLUSION

The JavaScript Grade Calculator project successfully demonstrates how client-side scripting can transform static HTML forms into responsive, automated analytical utilities. By utilizing vanilla JavaScript for arithmetic processing and conditional evaluation, the application executes rapid client-side calculations while eliminating unnecessary server round-trips.

The system's control architecture ensures robust grade distribution and verifies subject-specific criteria, maintaining academic assessment integrity through dual-layer evaluation: composite average banding for letter grades (*A +* through *F*) and explicit per-subject cutoff checks for pass/fail determination. Furthermore, state-driven DOM updates dynamically unhide and populate the results pane in real time.

Ultimately, this implementation emphasizes core software development practices, demonstrating clean form validation, event handling, conditional logic execution, and asynchronous-like client-side rendering using foundational web standards.